Maurizio Ungaro

Application Profile

Staff Scientist, Jefferson Lab | Nuclear physicist and simulation software developer

ungaro@jlab.org maureeungaro.github.io/home github.com/maureeungaro

Google Scholar INSPIRE ORCID 0000-0001-6982-3310 LinkedIn

Positioning Statement

I am a nuclear physicist and scientific software developer whose work connects physics analysis, detector systems, and production-scale simulation infrastructure. I am particularly interested in roles that value both scientific judgment and practical software delivery: simulation frameworks, detector operations, computational science, research software engineering, and technical leadership.

Value for Future Roles

- I can bridge domain science and production software: Geant4, GEMC, CLAS12 Simulations geometry, digitization, and analysis workflows.
- I have experience supporting users through tutorials, examples, technical notes, and operational guidance.
- I understand detector operations and calibration through CLAS12 Low Threshold Cherenkov Counter work, not only simulation abstractions.
- I have delivered infrastructure for distributed simulation production on HTCondor and Open Science Grid resources.
- I can communicate to multiple audiences: collaborators, students, detector experts, software users, and review committees.

Application Themes

Scientific computing and research software engineering: emphasize GEMC, Geant4, Python/C++, reproducibility, CI, Docker, and user support.

Detector and experimental physics: emphasize Low Threshold Cherenkov Counter, CLAS12 Simulations, Jefferson Lab Hall-B, calibration, detector performance, and nucleon-structure analyses.

Collaboration breadth: where relevant, mention prior work connected to SPring-8 and dark-sector searches through Heavy Photon Search.

Leadership and user enablement: emphasize tutorials, technical notes, simulation releases, OSG workflows, and cross-collaboration support.

Teaching and mentoring: emphasize undergraduate laboratory instruction, student supervision at Jefferson Lab, and practical mentoring in data analysis, hardware calibration, detector simulation, and scientific computing.

Reusable Short Cover Paragraph

My work at Jefferson Lab combines nuclear physics, detector systems, and scientific software. I develop and support Geant4/GEMC simulation workflows, maintain CLAS12 production tools, contribute to detector operation and calibration, and write documentation and tutorials that help users apply complex simulation tools productively. This combination of physics depth, software delivery, and user support is the foundation I would bring to a new role.

Reusable Teaching Paragraph

My teaching and mentoring experience comes from both formal undergraduate laboratory instruction and day-to-day research supervision. I taught PHYS 105 and PHYS 152 laboratories at Christopher Newport University, served as a teaching assistant for Physics I and Electromagnetic Theory at Rensselaer Polytechnic Institute, and have mentored students at Jefferson Lab on data analysis, hardware calibration, detector simulation, and physics analysis. My teaching style emphasizes practical problem solving, experimental judgment, and modern computing tools that help students connect physical concepts to measurable data.

Profiles

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